

Synthetic Methods

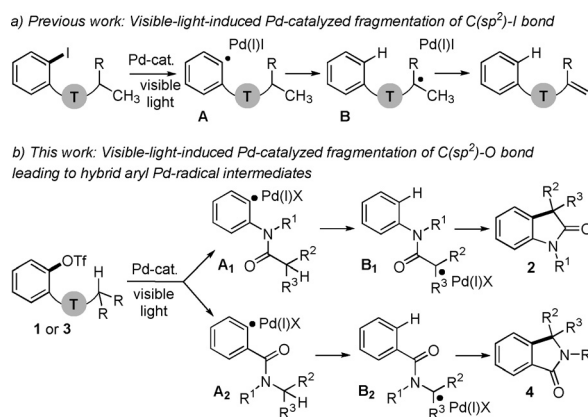
Visible-Light-Induced Palladium-Catalyzed Generation of Aryl Radicals from Aryl Triflates

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Abstract: A mild visible-light-induced Pd-catalyzed intramolecular C–H arylation of amides is reported. The method operates by cleavage of a C(sp²)–O bond, leading to hybrid aryl Pd-radical intermediates. The following 1,5-hydrogen atom translocation, intramolecular cyclization, and rearomatization steps lead to valuable oxindole and isoindoline-1-one motifs. Notably, this method provides access to products with readily enolizable functional groups that are incompatible with traditional Pd-catalyzed conditions.

Aryl halides are essential substrates for a variety of Pd-catalyzed transformations, which traditionally rely on the Pd^{II} intermediates formed by a two-electron oxidative-addition process.^[1] Recently, a new approach involving intermediacy of a Pd^I species generated by a single-electron-transfer (SET) step from a photoexcited Pd⁰ complex to an organohalide, emerged as a powerful and rapidly expanding area of Pd catalysis.^[2] Recently, our group reported visible-light-induced Pd-catalyzed cleavage of the C(sp²)–I bond of aryl iodides, leading to formation of aryl hybrid Pd-radical intermediates **A** (Scheme 1a).^[3] The latter underwent a selective 1,5-hydrogen atom translocation step (HAT)^[4] (**B**) followed by β-H elimination, leading to silyl enol ethers^[3] and desaturated amines^[5] in an efficient manner. This strategy was also successfully translated to the fragmentation of C(sp²)–I bonds of vinyl iodides, leading to the formation of vinyl hybrid Pd-radical intermediates.^[6] Notably, these novel methods^[7] rely on the employment of organic halides,^[8] predominantly iodides and bromides, which limits the synthetic applicability of this approach. Potential employment of alternative aryl electrophiles would substantially expand the variety of available substrates for these transformations. Herein, we report mild visible-light-induced^[9] palladium-catalyzed fragmentation of C(sp²)–O bonds of aryl trifluoromethanesulfonates (triflates) **1**,^[10] leading to the formation of aryl hybrid Pd-radicals **A**.^[11] The latter, upon sequential 1,5-HAT (**A**₁→**B**₁)^[12], intramolecular cyclization,

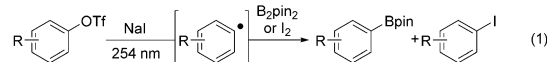
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Scheme 1. Aryl iodides and triflates in visible-light-induced Pd-catalyzed transformations. T = Tether.

and rearomatization steps, provided valuable oxindole products **2** (Scheme 1b).^[13]

Encouraged by the successful cleavage of aryl halides,^[3,5,14] and inspired by work from the Li group^[15] on UV-light-induced transition-metal-free fragmentation of (Csp²)–O bonds [Eq. (1)], we set to investigate the feasibility of generating hybrid aryl Pd-radical intermediates from aryl triflates under visible-light/Pd conditions.



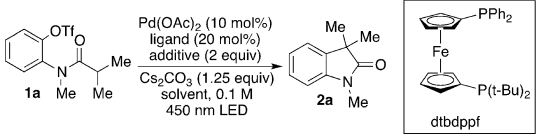
The intramolecular coupling of aryl triflates **1** was chosen for this study. Hartwig and co-workers^[16] reported intramolecular α-arylation of amides as a powerful approach towards the construction of oxindole cores.^[13b,c,d] Although high yielding and enantioselective, this method relies on the employment of strong bases, which notably diminishes the scope of this approach.

We began our study with examination of the intramolecular α-arylation of triflate **1a** (Table 1). Previously optimized reaction conditions proved to be inefficient (entry 1),^[3] however, we found that oxindole **2a** could be formed in low yield in *N,N*-dimethylformamide (DMF) when the bis[(2-diphenylphosphino)phenyl] ether (dpephos) ligand was employed. Further optimization revealed that this intramolecular arylation proceeds better in the presence of NaI (entry 3). Extensive ligand screening found 2,2'-bis(diphenylphosphino)-1,1'-binaphthalene (*rac*-BINAP) to be a superior ligand, leading to oxindole **2a** in 41 % yield. Screening other additives, known to affect Pd-catalyzed transformations of aryl triflates by stabilization of oxidative-addition adducts,^[17]

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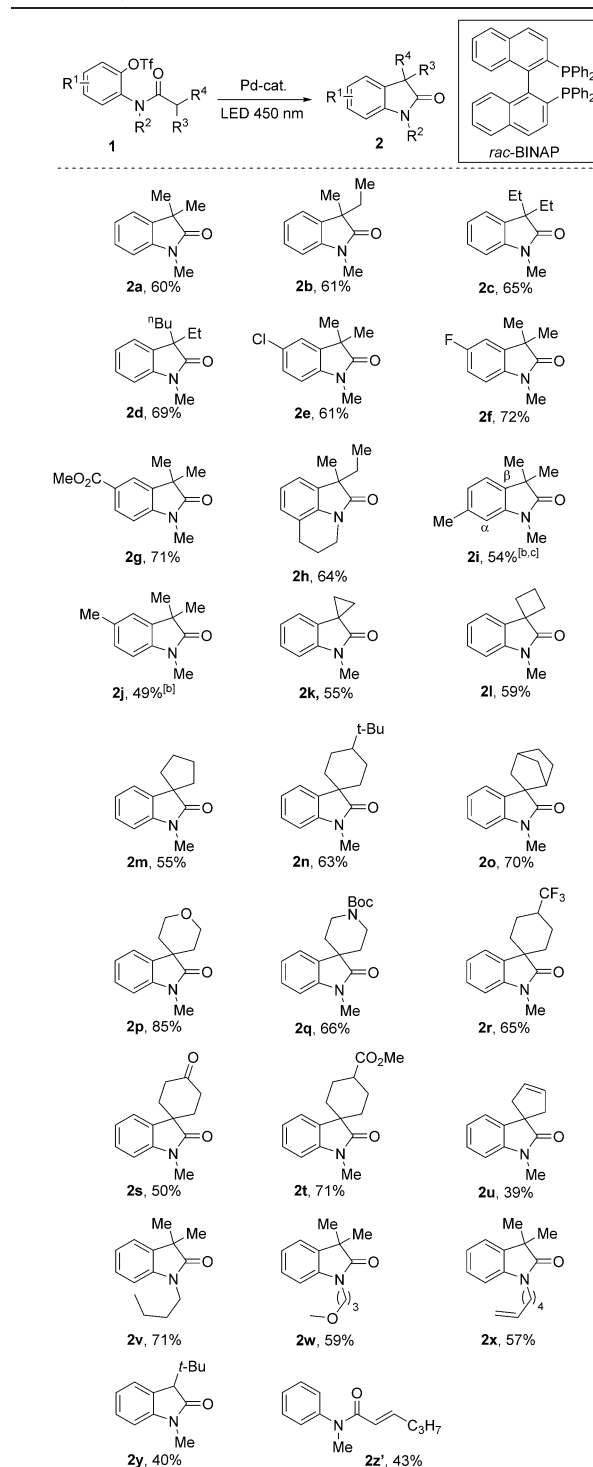
Table 1: Optimization of reaction parameters.^[a]


Entry	Ligand	Additive	Solvent	Yield [%] ^[b]
1 ^[c,d]	dtbdppf	–	benzene	0
2	dpephos	–	DMF	11
3	dpephos	NaI	DMF	30
4	<i>rac</i> -BINAP	NaI	DMF	41
5	<i>rac</i> -BINAP	NaCl	DMF	18
6	<i>rac</i> -BINAP	NaOAc	DMF	19
7	<i>rac</i> -BINAP	NaI	1,4-dioxane	62
8 ^[e]	<i>rac</i> -BINAP	NaI	1,4-dioxane	< 1
9 ^[f]	<i>rac</i> -BINAP	NaI	1,4-dioxane	< 1
10 ^[g]	<i>rac</i> -BINAP	NaI	1,4-dioxane	5

[a] All reactions were performed on a 0.1 mmol scale. [b] GC–MS yield. [c] 2 equivalents of Cs₂CO₃ were used. [d] 1-Diphenylphosphino-1'-(di-*tert*-butylphosphino)ferrocene. [e] No Pd(OAc)₂ was added. [f] No light, T = rt to 100°C. [g] 2 equivalents of TEMPO were added.

did not lead to an improvement in the reaction (entries 5 and 6). Optimization of the solvent indicated that this reaction proceeds best in 1,4-dioxane, leading to the oxindole **2a** in 62 % yield. Control experiments confirmed that catalyst, light, and additive are all essential for the success of this transformation (entries 8, 9). Additionally, performing the reaction in the presence of radical scavenger 2,2,6,6-tetramethylpiperidine 1-oxyl (TEMPO; entry 10) led to a substantially diminished yield, thus indicating the possible involvement of radical intermediates.

With the optimized conditions in hand, the scope of this intramolecular arylation reaction towards the oxindole core^[18] was studied (Table 2). Thus, reaction of triflate **1a** led to the oxindole **2a** in 60 % yield. Introduction of various alkyl groups next to α -carbon atom posed no problem, leading to oxindoles **2b–2d** in 61, 65, and 69 % yield, respectively. Substrates **1e–1g**, which have halo- and ester functional groups on the aromatic ring, reacted well to provide products **2e–2g** in good yields. Likewise, tricyclic oxindole **2h** was obtained in 64 % yield. Alkyl substituents on the aryl ring were also tolerated. The scope of this transformation could further be expanded to spirocyclic oxindole motifs. Thus, spiro cyclopropyl- (**2k**), cyclobutyl- (**2l**), cyclopentyl- (**2m**), and cyclohexyl- (**2n**) oxindoles were synthesized efficiently by using this protocol. Norbornane derivative **1o** reacted smoothly leading to **2o** in 70 % yield. Incorporation of heteroatoms did not alter the course of the reaction to give oxindoles **2p–2r** in 85, 65, and 66 % yield, respectively. Notably, base-sensitive functionalities, such as ketone and ester groups are well tolerated under these reaction conditions. Thus, oxindoles **2s** and **2t**, possessing enolizable groups and thus, unavailable via traditional Pd-catalyzed methods,^[13,16] were isolated in 50 and 71 % yield, respectively. The transformation of cyclopentene derivative **1u** led to the corresponding oxindole **2u**, albeit in moderate yield. Varying the substitution pattern at the N atom led to oxindoles **2v–2x** in 71, 59, and 57 % yield, respectively. Notably, this trans-

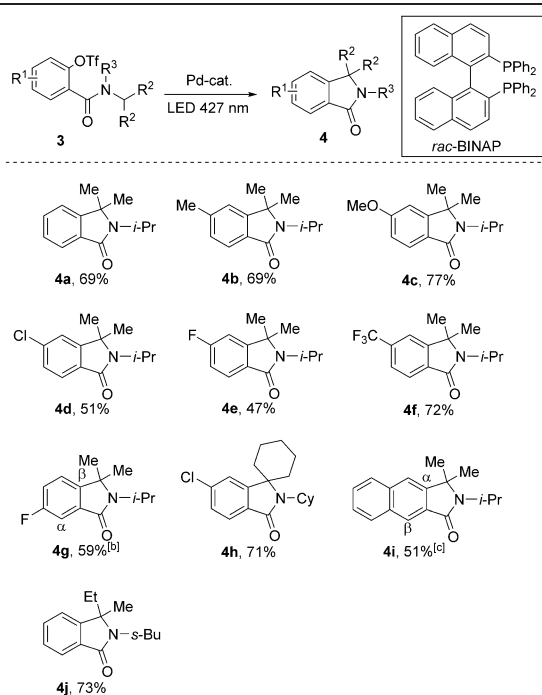
Table 2: Scope of oxindoles.^[a]

[a] Reaction conditions: **1** 0.1 mmol, Pd(OAc)₂ 0.01 mmol, *rac*-BINAP 0.02 mmol, NaI 0.2 mmol, Cs₂CO₃ 0.125 mmol, 1,4-dioxane 0.1 M, 34 W blue LED. [b] NMR yield. [c] α : β , 3:1.

formation was capable of delivering the 3-monosubstituted oxindole **2y**, though in a moderate 40 % yield. However, the replacement of the bulky *tert*-butyl group with the sterically less demanding *n*-pentyl moiety led to formation of acyclic desaturated amide **2z'** as the major reaction product.

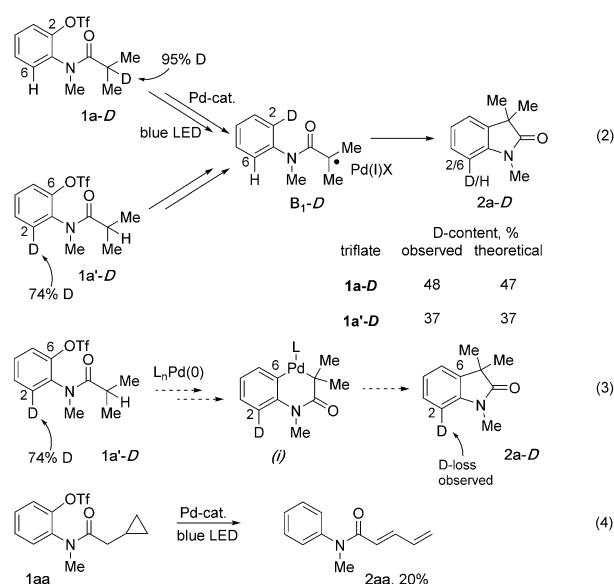
Notably, the developed method for the synthesis of oxindoles **2** operates under an electronically matched scenario, in which a translocated electrophilic radical **B₁** cyclizes at an electron-rich aromatic ring (Scheme 1b). We hypothesized that the analogous cyclization under an electronically inverted scenario (**A₂**→**B₂**) should be possible, thus leading to useful isoindoline-1-one products **4**.^[19] Therefore, the cyclization of benzamide^[20] **3a** was tested under the optimized reaction conditions. To our delight, isoindoline-1-one **4a** was isolated in 69% yield (Table 3). Reactions of differently substituted benzamides (**3b–g**, **3i**), featuring alkyl, methoxy, halo-, trifluoromethyl, and naphthyl groups proceeded smoothly, providing the corresponding isoindoline-1-ones in moderate-to-good yields. The protocol also proved to be efficient for the cyclization of cyclohexyl- and isobutyl amides, leading to products **4h** and **4j** in 71 and 73% yield.

Table 3: Scope of isoindoline-1-ones.^[a]



[a] Reaction conditions: **1** 0.1 mmol, Pd(OAc)₂ 0.01 mmol, *rac*-BINAP 0.02 mmol, NaI 0.2 mmol, Cs₂CO₃ 0.125 mmol, 1,4-dioxane 0.1 M, 40 W blue LED. [b] α:β, 2.5:1. [c] α:β, 2:1.

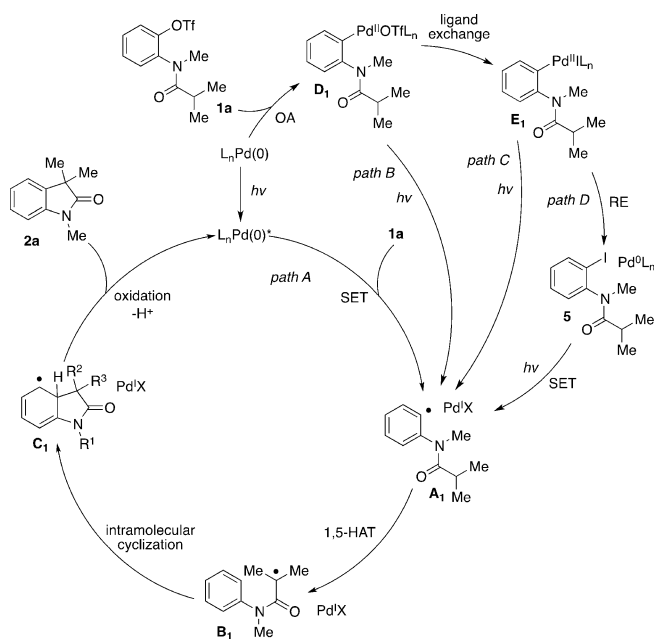
Next, we turned our attention to the mechanism of this intramolecular C–H arylation reaction. As mentioned above (Table 1, entry 10), the reaction was sensitive to the presence of a radical scavenger, leading to a dramatic decrease of the reaction yield. Additional evidence supporting the radical nature of this transformation was obtained by the following deuterium-labeling experiments [Eq. (2)]. Intramolecular arylation of deuterated triflate **1a-D** was investigated first. It was expected that upon fragmentation of the C–O bond and subsequent 1,5-HAT, the **B₁-D** intermediate would be produced. In the absence of any stereoelectronic factors, this intermediate was expected to display equal probability for



cyclization at either the C2- or C6-position of the aromatic ring, thus leading to the oxindole **2a-D** with a statistical preservation of the deuterium label at about 47%. Indeed, experiments with triflate **1a-D** completely supported the above projections, leading to the oxindole **2a-D** with 48% deuterium incorporation at the aromatic ring. Furthermore, cyclization of the alternatively **D**-labeled triflate, **1a'-D**, reacting via the same intermediate, **B₁-D**, was examined, and was found to produce the single reaction product oxindole **2a-D**. As in the previous experiment, an excellent agreement between the predicted and observed levels of deuterium-label incorporation was observed. These deuterium-labeling experiments ruled out a potential direct C-H activation mechanism (**1a'-D**→**i**→**2a-D**),^[21] according to which a complete preservation of the deuterium label at the C2-position should be observed [Eq. (3)].

In addition, the radical nature of this transformation was further confirmed by a radical-clock experiment with triflate **1aa** producing the ring-opened dienamide **2aa** [Eq. (4)], and by the spin-trap experiments with 5,5-dimethyl-1-pyrroline *N*-oxide (DMPO).^[22]

Based on the preliminary mechanistic studies, we postulate the formation of a key hybrid aryl Pd-radical intermediate **A₁** in this transformation (Scheme 2). A consecutive 1,5-HAT step produces the translocated hybrid alkyl radical **B₁**, which, upon intramolecular cyclization at the aromatic ring (**C₁**) and a subsequent rearomatization, is converted to the oxindole product **2a**. At present, the details of the C–O-bond-fragmentation step, as well as the precise role of NaI, remain unclear. We foresee the generation of intermediate **A₁** via several alternative scenarios. According to *path A*, a SET between aryl triflate **1a** and an active photoexcited Pd complex^[23] leads to the formation of an aryl triflate radical anion (not shown), which upon cleavage of the C–O bond leads to **A₁**. In *path B*, oxidative addition of the Pd catalyst onto **1a** leads to the adduct **D₁**, which then undergoes visible-light-induced homolysis of the C–Pd bond, thus leading to **A₁**. Alternatively, ligand exchange between **D₁** and NaI results in



Scheme 2. Proposed mechanism. OA = Oxidative addition; RE = reductive elimination.

a more stable aryl–Pd–I intermediate E_1 .^[17] As in the previous case, irradiation-induced homolysis of the C–Pd bond converts E_1 into A_1 (path C). Finally, a reductive elimination of the intermediate E_1 generates aryl iodide **5**. The latter undergoes SET with the photoexcited Pd catalyst,^[3,5] leading to hybrid aryl Pd-radical A_1 (path D). It should be mentioned that several additional mechanistic experiments, including an EPR study on model substrates, rate comparison of stoichiometric reactions with palladium, and monitoring parallel reactions at the early stages, suggest that all four scenarios are feasible.^[22] Accordingly, at this stage, none of the proposed pathways for fragmentation of the C–O bond may be reliably ruled out. Evidently, more detailed mechanistic studies are required to understand the precise mechanism for this novel transformation.

In summary, we have demonstrated that the combination of visible light and palladium catalysis can efficiently fragment the C(sp²)–O bond of readily available aryl triflates, leading to aryl hybrid Pd radicals. Involvement of these radicals in successive 1,5-HAT, cyclization, and rearomatization events afforded diversely substituted oxindole and isoindoline-1-one products. Advantageous engagement of a Pd⁰/Pd^I/Pd^{II} manifold over the traditional Pd⁰/Pd^{II} cycle allowed for the formation of oxindole products with base-sensitive functionalities that are incompatible with traditional Pd-catalyzed conditions. Although, the involvement of hybrid Pd radicals is well supported, the details of the C–O-bond-fragmentation step, as well as the role of NaI, remain unclear at this stage. More detailed mechanistic studies are required to elucidate the precise mechanism of this transformation. It is believed that this method will find applications in synthesis.

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Conflict of interest

The authors declare no conflict of interest.

Keywords: homolysis · light-mediated reactions · palladium · radicals · triflates

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